

Energy Through a Food Chain: The Efficiency Cascade

Name: _____

A worksheet on producers, trophic levels, and what happens when several imperfect energy transfers occur in sequence.

Starting idea. At each stage, only a fraction of the energy available at that stage becomes stored chemical energy in the next stage. The rest leaves the pathway through respiration, movement, heat, waste, uneaten biomass, and other losses. A useful ecological rule of thumb is that trophic transfer is often around 10%, but real values vary roughly from 5% to 20%. The accompanying “Efficiency Cascade” note gives this range and distinguishes trophic transfer from other kinds of efficiency.

1. Follow the energy, one stage at a time

A meadow receives **20,000 kJ** of solar energy over a chosen area and time interval.

Stage	Energy channel	Efficiency at this stage	Energy leaving stage / kJ
Producer	sunlight → grass biomass	1.8%	_____
Primary consumer	grass biomass → rabbit biomass	12%	_____
Secondary consumer	rabbit biomass → fox biomass	9%	_____

- a) For each row, calculate the energy leaving that stage. Be careful: each percentage acts on the energy **entering that row**, not on the original 20,000 kJ.
- b) What fraction of the original 20,000 kJ reaches fox biomass? _____
- c) Hence calculate the overall percentage efficiency from sunlight to fox biomass. _____

2. Three different pathways

Complete each pathway. Work from left to right.

A. Pond chain 15,000 kJ sunlight → algae (2.5%) → zooplankton (18%) → small fish (14%) → pike (7%).
Energy after algae: _____ kJ after zooplankton: _____ kJ
Energy after small fish: _____ kJ after pike: _____ kJ
Overall efficiency: _____%

B. Crop eaten directly 10,000 kJ sunlight → crop biomass (2.2%) → human biomass (25%).
Energy in crop biomass: _____ kJ energy reaching human biomass: _____ kJ
Overall efficiency: _____%

C. Woodland chain 30,000 kJ sunlight → leaves (1.2%) → caterpillar (15%) → songbird (11%) → hawk (8%).
Energy after leaves: _____ kJ caterpillar: _____ kJ songbird: _____ kJ hawk: _____ kJ
Overall efficiency: _____%

3. What operation combines efficiencies?

For the meadow chain, the three stage efficiencies were 1.8%, 12%, and 9%.

- Try **adding** the three percentages. What do you get? _____%
Does this equal your overall efficiency from Question 1? _____
- Write the three efficiencies as decimals: _____, _____, _____.
- Find a way of combining those three decimals that gives the overall decimal efficiency you found in Question 1.
Show your trial here:
- Complete the general rule you think is operating:

$$\eta_{\text{total}} = \boxed{}$$

where each stage efficiency is written as a decimal fraction.

- Explain in one sentence why simply adding successive efficiencies cannot work.

4. Test your rule without tracking every intermediate energy

Use your rule from Section 3 to calculate the total efficiency directly.

Pathway	Stage efficiencies	Overall efficiency
Grass → vole → owl	8%, 13%	_____
Phytoplankton → krill → penguin → seal	16%, 11%, 9%	_____
Maize → insect → chicken → human	7%, 18%, 12%	_____

5. Reverse the cascade

A food chain has three successive trophic-transfer efficiencies. The first is 10%, the second is 15%, and the overall efficiency is 0.12%.

- Convert 0.12% to a decimal fraction. _____
- Find the efficiency of the third transfer. _____%
- A student says: “Adding another trophic level always removes 90% of the *original* energy.” Explain precisely what is wrong with this statement.

6. Interpretation

- Which of the pathways in Section 2 has the greatest overall efficiency? _____
- Why can a short food chain have a large energetic advantage even if its individual transfer efficiencies are not especially high?
- The accompanying note describes trophic efficiency as an emergent statistical average rather than a fixed physical constant. Give two ecological reasons why the transfer percentage could vary from one trophic step to another.

Exit statement. Complete this without looking back: “Successive efficiencies combine by _____ because each percentage acts on _____.”